TENDER SPECIFICATION FOR

DOOR & ROLLING SHUTTER

FOR CCL-3 BUILDING

AT

JSW COATED STEEL PRODUCTS LIMITED

KALMESHWAR

Rev-1 on 18-10-2019

Doors & Rolling Shutter Requirements :-

Doors & Rolling Shutter Details					
S.N.	Type of Door / Shutter	Description	Door Size {M (W) x M (H)}	Location	Qty. (Nos)
1	VLSD-1	Motorized Vertical Lift Sectional Door	7 m (W) x 6 m (H)	At the Gable End of CCL-3	4
2	VLSD-2	Motorized Vertical Lift Sectional Door	6 m (W) x 5 m (H)	At the Gable End of Galvalume Extended Shed	2
3	VLSD-3	Motorized Vertical Lift Sectional Door **(Auto Opn.)**	4 m (W) x 4 m (H)	Along the F-Row For Coil Transfer Trolley.	1
4	VLSD-4	Motorized Vertical Lift Sectional Door	4 m (W) x 4 m (H)	Along the Grid-EF-1 For Guard Film Storage Area in **Brick Work.**	1
5	HSD-1	Motorized Horizontal Sliding Door	7 m (W) x 6 m (H)	Along the H-Row For Material Dispatch.	4
6	HSD-2	Motorized Horizontal Sliding Door	3 m (W) x 3 m (H)	Along the F-Row (2-2B), Guard Film Storage	1
7	RS-1	Single Wall Motorized Rolling Shutter	7 m (W) x 4.5 m (H)	Along E-Row (14-14A) - Truck Passage in **Brick Work**	1
8	RS-2	Single Wall Motorized Rolling Shutter	4 m (W) x 4 m (H)	Along F-Row (14-14A) - Truck Passage.	1
9	RS-3	Single Wall Motorized Rolling Shutter	4 m (W) x 6 m (H)	Along E-Row (4-5), (6-7) - RTO Room	2
10	RS-4	Single Wall Motorized Rolling Shutter	4 m (W) x 4.5 m (H)	Along E-Row (15A-16)- Paint Room in **Brick Work.**	1
11	RS-5	Single Wall Motorized Rolling Shutter **with Grill**	5 m (W) x 4.5 m (H)	Along E-Row (10-10A), (10A-11), (11-11A) - X^{mer} Room in **Brick Work**	3
		Total			**21**

Sr. No.-1:- Technical Specification For :- VLSD-1 (Qty.-4 Nos)

S.N.	Particulars	JSW Requirement	Remark
1	**General Technical Spec.**		
1.1	Door Type (VLSD-1)	Vertical Lift Overhead Sectional Doors	Yes
1.2	**VLSD-1** :- Size 7M (W) X 6.0M (H) at the Gable End of CCL-3	4 Nos.	Yes
1.3	**VLSD-1** :- Mounting Space	**200 mm Gap** for vertical lift of panels above the door height from 6 M to 9 M Height.	Required 550 mm
1.4	Opening Direction	Vertical	Yes
1.5	Mounting Location	Inside	Yes
1.6	Mounting Surface	Steel Frame Construction.	Yes
1.7	Wind Resistance	Class 3- Classification as per EN 12424	Class 3 EN21424
1.8	Air permeability	Class 2- Classification as per EN 12426	Yes
1.9	Resistance to water penetration	Class 3- Classification as per EN 12425	Yes
1.10	Heat Insulation	U = 1.1 W/m²k (complete door without wicket door)	Yes
1.11	Acoustics	25dB	Yes
2	**Details of Door Construction & Structure**		
2.1	Door Construction	Self-Supporting	Yes
2.2	Door Panel (PUF Insulated)	Finger trap safe joint panel	Yes
		0.50 mm thk GI pre coated sheet in-filled with **CFC Free** (Chloro Fluoro Carbide) self-extinguishing **40/42 mm** thk Polyurethane foam.	Yes
		PUF Density **40-45 Kg/M³**	Yes
2.3	Height of door panel & No. of Door panel in each door	As per Design {Approximately 10 Panel (6000/600)}	545 mm

2.4	Side Guide Frame Profile	Profiled hot-dip galvanized steel frame of **Min. 2.00 mm thick.** having protective strip with **EPDM** (Ethylene Propylene Diene Monometer) Sealing on both sides.	Yes
2.5	Bottom Profile	Safety Bottom profile of Galvanized Steel of min **2.0 mm** thickness with EPDM seal	Yes
2.6	Hinges	Galvanized Steel hinges at sides with roller carrier & Galvanized Steel hinges at center.	Yes
2.7	Roller	Precisely made lateral rollers with ball bearings	Yes
2.8	Fittings	Vertical Lift Fitting	Yes
2.9	Roller Shaft & Bracket	As per Supplier's Design: - Torsion spring shaft with lateral load-bearing cables & Heavy duty steel Bracket. (Supplier has to specify the size & thickness)	Yes
2.10	Sealing	EPDM upper header seal, floor seal & center seal.	Yes
2.11	Wicket Door	**Wicket Door Require in Each Door (Approx. Size-1 m x 2.2 m**	Yes : 0.9m x 2.2m
2.12	Vision Panel	**5 Nos/Door** (At 1.2 m height from bottom), Glazing: Rectangular glazing constructed with single or double glazed panels. Approx. Vision Panel Size: 600-650 mm (W) x 300-350 mm (H)	Yes
2.13	Surface Finish & Color	Outside : Stucco **(RAL 3016)** Finish; Inside : Stucco **(RAL 9010)** finish.	Yes
3	**Details of Automation & Accessories**		
3.1	Control Type	Pulse Control with dead man control	Pluse control but completely automatic Just press the operator button once & the door will no function to set opening limit
3.2	Drive Unit	Direct Drive easy to mount with in built Limit Switches. **Supplier has to specify the drive details.**	Yes
3.3	No of cycle per Hour	20 cycles in per hour	Yes
3.4	Operating Switch	Three Station Push Button :- **2 Nos.** for Inside & Outside	Yes
3.5	Opening/Closing Speed	0.12 – 0.20 mtr / sec	0.25m/s

3.6	Emergency Operation	Manual Hand Chain	OK
3.7	Safety Catch for door (Torsion spring breakage protection device)	Breaking device to avoid falling by self wt.	Yes
3.8	Safety Side trap guard	For Cable & Manual intervention.	Yes
3.9	Mechanical / Operational / Safety Accessories	Motor Cover: GI Sheet of Min. 0.8mm thick	Yes
		Bottom Rubber Sealing	Yes
		Fixed Bollards (4 Nos.per Door of 800 mm height)	Yes
		Reflective/Indicative strips on guides/bottom profile	Yes
		Key Switch	Yes
		Flashing Light with Alarm	Yes
		Traffic Light, Red for Closed & Green for Open on both side (Inside &Outside)	Yes
		Safety Photocell (Transmitter & Receiver)	Yes
		Optical Wireless safety edge sensor	Yes wired safety edge Sensor, because need to change battery every 15 days
4	**Spare List for Doors**		
4.1	Controller box (including Card and Push button)	2 No	Yes
4.2	Complete Rollers wheel set assembly of gate	4 Set	Yes
4.3	Photo cell pair (Transmitter & Receiver)	1 Set	Yes
4.4	Special cable (From Controller to sensor)	1 Set	
4.5	Wireless safety edge sensor	1 Set	Yes wired safety edge Sensor ,because need to change battery every 15 days

Sr. No.-2 :- Technical Specification For :- VLSD-2 (Qty.-2 Nos)

S.N.	Particulars	JSW Requirement	Remark
1	**General Technical Spec.**		
1.1	Door Type (VLSD-2)	Vertical Lift Overhead Sectional Doors	Yes
1.2	**VLSD-2** :- Size 6.0M (W) X 5.0M (H) at the Galvalume Extended Shed	2 Nos.	Yes
1.3	**VLSD-2** :- Mounting Space	**200 mm Gap** for vertical lift of panels above the door height from 5 M to 9 M Height.	Require 550mm
1.4	Opening Direction	Vertical	Yes
1.5	Mounting Location	Inside	Yes
1.6	Mounting Surface	Steel Frame Construction.	Yes
1.7	Wind Resistance	Class 3- Classification as per EN 12424	Class 3 EN21424
1.8	Air permeability	Class 2- Classification as per EN 12426	Yes
1.9	Resistance to water penetration	Class 3- Classification as per EN 12425	Yes
1.10	Heat Insulation	U = 1.1 W/m²k (complete door without wicket door)	Yes
1.11	Acoustics	25dB	Yes
2	Details of Door Construction & Structure		
2.1	Door Construction	Self Supporting	Yes
2.2	Door Panel (PUF Insulated)	Finger trap safe joint panel.	Yes
		0.50 mm thk GI pre coated sheet in-filled with **CFC Free** (Chloro Fluoro Carbide) self extinguishing **40/42 mm** thk Polyurethane foam.	Yes
		PUF Density **40-45 Kg/M^3**	Yes
2.3	Height of door panel & No. of Door panel in each door	As per Design {Approximately 8 Panel (5000/600)}	545mm
2.4	Side Guide Frame Profile	Profiled hot-dip galvanized steel frame of **Min. 2.00 mm thick.** having protective strip with **EPDM** (Ethylene	Yes

		Propylene Diene Monometer) Sealing on both sides.	
2.5	Bottom Profile	Safety Bottom profile of Galvanized Steel of min **2.0 mm** thickness with EPDM seal	Yes
2.6	Hinges	Galvanized Steel hinges at sides with roller carrier & Galvanized Steel hinges at center.	Yes
2.7	Roller	Precisely made lateral rollers with ball bearings	Yes
2.8	Fittings	Vertical Lift Fitting	Yes
2.9	Roller Shaft & Bracket	As per Supplier's Design :- Torsion spring shaft with lateral load-bearing cables & Heavy duty steel Bracket. (Supplier has to specify the size & thickness)	Yes
2.10	Sealing	EPDM upper header seal, floor seal & center seal.	Yes
2.11	Wicket Door	**Wicket Door Require in Each Door (Approx. Size-1 m x 2.2 m**	Yes0.9m x 2.2m
2.12	Vision Panel	**3 to 4 Nos/Door** (At 1.2 m height from bottom), Glazing: Rectangular glazing constructed with single or double glazed panels. Approx. Vision Panel Size: 600-650 mm (W) x 300-350 mm (H)	Yes
2.13	Surface Finish & Color	Outside : Stucco **(RAL 3016)** Finish; Inside : Stucco **(RAL 9010)** finish.	Yes
3	**Details of Automation & Accessories**		
3.1	Control Type	Pulse Control with dead man control	Pluse control but completely automatic Just press the operator button once & the door will no function to set opening limit
3.2	Drive Unit	Direct Drive easy to mount with in built Limit Switches. **Supplier has to specify the drive details.**	Yes
3.3	No of cycle per Hour	20 cycles in per hour	Yes
3.4	Operating Switch	Three Station Push Button :- **2 Nos.** for Inside & Outside	Yes
3.5	Opening/Closing Speed	0.12 – 0.20 mtr / sec	0.25m/s
3.6	Emergency Operation	Manual Hand Chain	yes
3.7	Safety Catch for door (Torsion spring breakage protection device)	Breaking device to avoid falling by self wt.	Yes

3.8	Safety Side trap guard	For Cable & Manual intervention.	Yes
3.9	Mechanical / Operational / Safety Accessories	Motor Cover: GI Sheet of Min. 0.8mm thick	Yes
		Bottom Rubber Sealing	Yes
		Fixed Bollards (4 Nos.per Door of 800 mm height)	Yes
		Reflective/Indicative strips on guides/bottom profile	Yes
		Key Switch	Yes
		Flashing Light with Alarm	Yes
		Traffic Light, Red for Closed & Green for Open on both side (Inside &Outside)	Yes
		Safety Photocell (Transmitter & Receiver)	Yes
		Optical Wireless safety edge sensor	Yes wired safety edge Sensor, because need to change battery every 15 days
4	**Note :- All the Accessories of VLSD-2 to be standardized with VLSD-1**		Yes

Sr. No.-3 :- Technical Specification For :- VLSD-3 (Qty.-1 Nos)

S.N.	Particulars	JSW Requirement	Remark
1	**General Technical Spec.**		
1.1	Door Type (VLSD-3)	Vertical Lift Overhead Sectional Doors	Yes
1.2	**VLSD-3 :-** Size **4M (W) x 4M (H)** along the F-Row For Coil Transfer Trolley.	1 No	Yes

1.3	**VLSD-3** :- Mounting Space	Straight Vertical lift upto 2.5 m Height from Top of the door then horizontal move upto 1.5 M.	Yes
1.4	Opening Direction	Vertical	High Lift
1.5	Mounting Location	Inside	Yes
1.6	Mounting Surface	Steel Frame Construction.	Yes
1.7	Wind Resistance	Class 3- Classification as per EN 12424	Class 3 EN21424
1.8	Air permeability	Class 2- Classification as per EN 12426	Yes
1.9	Resistance to water penetration	Class 3- Classification as per EN 12425	Yes
1.10	Heat Insulation	U = 1.1 W/m²k (complete door without wicket door)	Yes
1.11	Acoustics	25dB	Yes
2	**Details of Door Construction & Structure**		
2.1	Door Construction	Self Supporting	Yes
2.2	Door Panel (PUF Insulated)	Finger trap safe joint panel	Yes
		0.50 mm thk GI pre coated sheet in-filled with **CFC Free** (Chloro Fluoro Carbide) self extinguishing **40/42 mm thk** Polyurethane foam.	Yes
		PUF Density **40-45 Kg/M^3**	Yes
2.3	Height of door panel & No. of Door panel in each door	As per Design {Approximately 7 Panel (4000/600)}	545mm
2.4	Side Guide Frame Profile	Profiled hot-dip galvanized steel frame of **Min. 2.00 mm thick.** having protective strip with **EPDM** (Ethylene Propylene Diene Monometer) Sealing on both sides.	Yes
2.5	Bottom Profile	Safety Bottom profile of Galvanized Steel of **min 2.0 mm** thickness with EPDM seal	Yes
2.6	Hinges	Galvanized Steel hinges at sides with roller carrier & Galvanized Steel hinges at center.	Yes
2.7	Roller	Precisely made lateral rollers with ball bearings	Yes
2.8	Fittings	Vertical Lift Fitting upto **2.5 m Height from Top** of the door then **horizontal move upto 1.5 M.**	Yes
2.9	Roller Shaft & Bracket	As per Supplier's Design :- Torsion spring shaft with lateral load-bearing cables & Heavy duty steel Bracket.	Yes

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		(Supplier has to specify the size & thickness)	
2.10	Sealing	EPDM upper header seal, floor seal & center seal.	Yes
2.11	Wicket Door	**Wicket Door Require** (Approx. Size-1 m x 2.2 m)	Yes0.9m x 2.2m
2.12	Vision Panel	**3 Nos/Door** (At 1.2 m height from bottom), Glazing: Rectangular glazing constructed with single or double glazed panels. Approx. Vision Panel Size: 600-650 mm (W) x 300-350 mm (H)	Yes
2.13	Surface Finish & Color	Outside : Stucco **(RAL 3016)** Finish; Inside : Stucco **(RAL 9010)** finish.	Yes
3	**Details of Automation & Accessories**		
3.1	Control Type	Pulse Control with dead man control	Pluse control but completely automatic Just press the operator button once & the door will no function to set opening limit
3.2	Drive Unit	Direct Drive easy to mount with in built Limit Switches. **Supplier has to specify the drive details.**	Yes
3.3	No of cycle per Hour	20 cycles in per hour	Yes
3.4	Operating Switch	Three Station Push Button :- **2 Nos**. for Inside & Outside	Yes
3.5	Opening/Closing Speed	0.12 – 0.20 mtr / sec	0.25m/s
3.6	Emergency Operation	Manual Hand Chain	Yes
3.7	Safety Catch for door (Torsion spring breakage protection device)	Breaking device to avoid falling by self wt.	Yes
3.8	Safety Side trap guard	For Cable & Manual intervention.	Yes
3.9	Mechanical / Operational / Safety Accessories	Motor Cover: GI Sheet of Min. 0.8mm thick	Yes
		Bottom Rubber Sealing	Yes
		Fixed Bollards (4 Nos.per Door of 800 mm height)	Yes
		Reflective/Indicative strips on guides/bottom profile	Yes
		Key Switch	Yes
		Flashing Light with Alarm	Yes
		Traffic Light, Red for Closed & Green for Open on both side (Inside	Yes

		&Outside)	
		Safety Photocell (Transmitter & Receiver)	Yes
		Optical Wireless safety edge sensor	Yes wired safety edge sensor ,because need to change battery every 15 days
3.9	**Auto Operation**	**Radar on Both Side**	Yes
4	**Note :- All the Accessories of VLSD-3 to be standardized with VLSD-1**		Yes

Sr. No.-4 :- Technical Specification For :- VLSD-4 (Qty.-1 Nos)

S.N.	Particulars	JSW Requirement	Remark
1	**General Technical Spec.**		
1.1	Door Type (VLSD-4)	Vertical Lift Overhead Sectional Doors	Yes
1.2	**VLSD-4 :-** Size **4M (W) x 4M (H)** along the Grid-EF-1 For Guard Film Storage Area.	1 No	Yes
1.3	**VLSD-4 :-** Mounting Space	Horizontal below Building Roof from Top of the door	Require 550mm
1.4	Opening Direction	Verical with Horizontal Fitting at Roof	Yes
1.5	Mounting Location	Inside	Yes
1.6	Mounting Surface	On Brick Work	Yes
1.7	Wind Resistance	Class 3- Classification as per EN 12424	Class 3 EN21424
1.8	Air permeability	Class 2- Classification as per EN 12426	Yes
1.9	Resistance to water penetration	Class 3- Classification as per EN 12425	Yes
1.10	Heat Insulation	U = 1.1 W/m²k (complete door without wicket door)	Yes

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1.11	Acoustics	25dB	Yes
2	**Details of Door Construction & Structure**		
2.1	Door Construction	Self Supporting	Yes
2.2	Door Panel (PUF Insulated)	Finger trap safe joint panel	Yes
		0.50 mm thk GI pre coated sheet in-filled with **CFC Free** (Chloro Fluoro Carbide) self extinguishing **40/42 mm** thk Polyurethane foam.	Yes
		PUF Density **40-45 Kg/M^3**	Yes
2.3	Height of door panel & No. of Door panel in each door	As per Design {Approximately 7 Panel (4000/600)}	545mm
2.4	Side Guide Frame Profile	Profiled hot-dip galvanized steel frame of **Min. 2.00 mm thick.** having protective strip with **EPDM** (Ethylene Propylene Diene Monometer) Sealing on both sides.	Yes
2.5	Bottom Profile	Safety Bottom profile of Galvanized Steel of min **2.0 mm** thickness with EPDM seal	Yes
2.6	Hinges	Galvanized Steel hinges at sides with roller carrier & Galvanized Steel hinges at center.	Yes
2.7	Roller	Precisely made lateral rollers with ball bearings	Yes
2.8	Fittings	Horizontal Lift Fitting below building Roof	Yes
2.9	Roller Shaft & Bracket	As per Supplier's Design :- Torsion spring shaft with lateral load-bearing cables & Heavy duty steel Bracket. (Supplier has to specify the size & thickness)	Yes
2.10	Sealing	EPDM upper header seal, floor seal & center seal.	Yes
2.11	Vision Panel	**3 Nos/Door (** At 1.2 m height from bottom), Glazing: Rectangular glazing constructed with single or double glazed panels. Approx Vision Panel Size: 600-650 mm (W) x 300-350 mm (H)	Yes
2.12	Surface Finish & Color	Outside : Stucco **(RAL 3016)** Finish; Inside : Stucco **(RAL 9010)** finish.	Yes
3	**Details of Automation & Accessories**		
3.1	Control Type	Pulse Control with dead man control	Pluse control but completely automatic Just press the operator button once & the door

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			will no function to set opening limit
3.2	Drive Unit	Direct Drive easy to mount with in built Limit Switches. Supplier has to specifty the drive details.	Yes
3.3	No of cycle per Hour	20 cycles in per hour	Yes
3.4	Operating Switch	Three Station Push Button :- **2 Nos.** for Inside	Yes
3.5	Opening/Closing Speed	0.12 – 0.20 mtr / sec	0.25m/s
3.6	Emergency Operation	Manual Hand Chain	Ok
3.7	Safety Catch for door (Torsion spring breakage protection device)	Breaking device to avoid falling by self wt.	Yes
3.8	Safety Side trap guard	For Cable & Manual intervention.	Yes
3.9	Mechanical / Operational / Safety Accessories	Motor Cover: GI Sheet of Min. 0.8mm thick	Yes
		Bottom Rubber Sealing	Yes
		Fixed Bollards (4 Nos.per Door of 800 mm height)	Yes
		Reflective/Indicative strips on guides/bottom profile	Yes
		Key Switch	Yes
		Flashing Light with Alarm	Yes
		Traffic Light, Red for Closed & Green for Open on both side (Inside &Outside)	Yes
		Safety Photocell (Transmitter & Receiver)	Yes
		Optical Wireless safety edge sensor	Yes wired safety edge, because need to change battery every 15 days
4	**Note :- All the Accessories of VLSD-4 to be standardized with VLSD-1**		Yes

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Sr. No.-5 :- Technical Specification For :- HSD-1 (Qty.-4 Nos)

S.N.	Particulars	JSW Requirement	Remark
1	**General Technical Spec.**		
1.1	Door Type **(HSD-1)**	Motorized Horizontal Sliding Doors with four leaf telescopic panel & bilateral movement.	Yes
1.2	**HSD-1** :- Size **7M (W) X 6M (H)** along the H-Row For Material Dispatch.	4 No.	Yes
1.3	Opening Direction	Horizontal (Bilateral-Telescopic)	Yes
1.4	Mounting Location	Outside (On **Top / Bottom** Track Mechanism)	Yes
1.5	Mounting Surface	Steel Frame	Yes
1.6	Wind Resistance	Class 3- Classification as per EN 12424	Yes
1.7	Air permeability	Class 2- Classification as per EN 12426	Yes
1.8	Resistance to water penetration	Class 3- Classification as per EN 12425	Yes
1.9	Heat Insulation	U = 1.1 W/m²k (complete door without wicket door)	Yes
1.10	Acoustics	25dB	Yes
2	**Details of Door Construction & Structure**		
2.1	Door Construction	Self Supporting	Yes
2.2	Door Panel (PUF Insulated)	Finger trap safe joint panel	Yes
		0.50 mm thk GI pre coated sheet in-filled with **CFC free** (Chloro Fluoro Carbide) self extinguishing **40/42 mm thk** Polyurethane foam.	Yes

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		PUF Density **40-45 Kg/M^3**	**40Kg/M^3**
2.3	No of Door Panels	4 Nos. **(2+2) (Telescopic)**	Yes
2.4	Door Outer Frame	G.I "C" type profile frame for all type of Cladding Sheet with painting.	Yes
2.5	Top Track with Bracket	GI Precoated / MS Plate with proper painting, top track for sliding Rollers with end stopper.	Yes
2.6	Bottom Track	As per design **(Ensuring the clear passage for truck movement)**	
2.7	Rollers	Top Rollers : Dual Type Top Rollers assembly with bearings. Bottom Rollers: Pin type Bottom Rollers assembly with bearings for guiding.	Yes
2.8	Sealing	EPDM Rubber Sealing for Side Lap other area	Yes
2.9	Door Operation Mechanism	**Bottom Mounted Rack & pinion** Mechanism **(Ensuring the clear passage for truck movement)**	Yes
2.10	Vision Panel	**4 Nos (** At 1.2 m height from bottom), Glazing: Rectangular glazing constructed with single or double glazed panels. Approx. Vision Panel Size: 600-650 mm (W) x 300-350 mm (H)	Yes
2.11	Surface Finish & Color	Outside : Stucco **(RAL 3016)** Finish; Inside : Stucco **(RAL 9010)** finish.	Yes
3	**Details of Automation & Accessories**		
3.1	Control Type	Pulse Control with dead man control	Yes
3.2	Motor Mounting	Top Mounted Rack Driven rugged (for clear space at bottom for material movement)	Yes
3.3	Drive Type	Reversible gear Motor with Rack	Yes
3.4	Metal Rack	Zinc platted Metal Rack	Yes
3.5	Operating Switch	Three Station Push Button :- **2 Nos.** for Inside	Yes
3.6	Opening/Closing Speed	0.12 – 0.20 mtr / sec	Yes
3.7	Manual Operation	Easy to operate with Release Device	Yes
3.8	No of cycle per Hour	20 cycles in per hour	Yes
3.9	Mechanical / Operational / Safety Accessories	Motor Cover: GI Sheet of Min. 0.8mm thick	Yes
		Bottom Rubber Sealing	Yes

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		Fixed Bollards (4 Nos.per Door of 800 mm height)	Yes
		Reflective/Indicative strips on guides/bottom profile	Yes
		Key Switch	Yes
		Flashing Light with Alarm	Yes
		Traffic Light, Red for Closed & Green for Open on both side (Inside &Outside)	Yes
		Safety Photocell (Transmitter & Receiver)	Yes
		Optical Wireless Vertical safety edge sensor	Yes
4	**Spare List for Doors**		
4.1	Controller box (including Card and Push button)	2 No	Yes
4.2	Complete set Pinion	1 Set	Yes
4.3	Photo cell pair (Transmitter & Receiver)	1 Set	Yes
4.4	Special cable (From Controller to sensor)	1 Set	Yes
4.5	Wireless safety edge sensor	1 Set	Yes

Sr. No.-6 :- Technical Specification For :- HSD-2 (Qty.-1 Nos)

S.N.	Particulars	JSW Requirement	Remark
1	**General Technical Spec.**		
1.1	Door Type **(HSD-2)**	Motorized Horizontal Sliding Doors with Double leaf & bilateral movement.	Yes
1.2	**HSD-2** :- Size **3M (W) X 3M (H)** along the F-Row (2-2B), Guard Film Storage	1 No.	Yes
1.3	Opening Direction	Horizontal (Bilateral)	Yes
1.4	Mounting Location	Outside (On Top Track Mechanism)	Yes

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1.5	Mounting Surface	Steel Frame	Yes
1.6	Wind Resistance	Class 3- Classification as per EN 12424	Yes
1.7	Air permeability	Class 2- Classification as per EN 12426	Yes
1.8	Resistance to water penetration	Class 3- Classification as per EN 12425	Yes
1.9	Heat Insulation	U = 1.1 W/m²k (complete door without wicket door)	Yes
1.10	Acoustics	25dB	Yes
2	**Details of Door Construction & Structure**		
2.1	Door Construction	Self Supporting	Yes
2.2	Door Panel (PUF Insulated)	Finger trap safe joint panel	Yes
		0.50 mm thk GI pre coated sheet in-filled with **CFC Free** (Chloro Fluoro Carbide) self extinguishing **40/42 mm thk** Polyurethane foam.	Yes
		PUF Density **40-45 Kg/M^3**	Yes
2.3	No of Door Panels	2 Nos. **(1 +1)**	Yes
2.4	Door Outer Frame	G.I "C" type profile frame for all type of Cladding Sheet with painting.	Yes
2.5	Top Track with Bracket	GI Precoated / MS Plate with proper painting, top track for sliding Rollers with end stopper.	Yes
2.6	Bottom Track	As per design **(Ensuring the clear passage for truck movement)**	Yes
2.7	Rollers	Top Rollers : Dual Type Top Rollers assembly with bearings. Bottom Rollers: Pin type Bottom Rollers assembly with bearings for guiding.	Yes
2.8	Sealing	EPDM Rubber Sealing for Side Lap other area	Yes
2.9	Door Operation Mechanism	**Top Mounted Rack & pinion** Mechanism **(Ensuring the clear passage for Material movement)**	Yes
2.10	Vision Panel	**2 Nos (** At 1.2 m height from bottom), Glazing: Rectangular glazing constructed with single or double glazed panels. Approx. Vision Panel Size: 600-650 mm (W) x 300-350 mm (H)	Yes
2.11	Surface Finish & Color	Outside : Stucco **(RAL-9010)** Finish; Inside : Stucco **(RAL 9010)** finish.	Yes
3	**Details of Automation & Accessories**		

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3.1	Control Type	Pulse Control with dead man control	Yes
3.2	Motor Mounting	Top Mounted Rack Driven rugged (for clear space at bottom for material movement)	Yes
3.3	Drive Type	Reversible gear Motor with Rack	Yes
3.4	Metal Rack	Zinc platted Metal Rack	Yes
3.5	Operating Switch	Three Station Push Button :- **2 Nos.** for Inside	Yes
3.6	Opening/Closing Speed	0.12 – 0.20 mtr / sec	Yes
3.7	Manual Operation	Easy to operate with Release Device	Yes
3.8	No of cycle per Hour	20 cycles in per hour	Yes
3.9	Mechanical / Operational / Safety Accessories	Motor Cover: GI Sheet of Min. 0.8mm thick	Yes
		Bottom Rubber Sealing	Yes
		Fixed Bollards (4 Nos.per Door of 800 mm height)	Yes
		Reflective/Indicative strips on guides/bottom profile	Yes
		Key Switch	Yes
		Flashing Light with Alarm	Yes
		Traffic Light, Red for Closed & Green for Open on both side (Inside &Outside)	Yes
		Safety Photocell (Transmitter & Receiver)	Yes
		Optical Wireless Vertical safety edge sensor	Yes
4	**Note :- All the Accessories of HSD-2 to be standardized with HSD-1**		Yes

Sr. No.-7,8,9 & 10 :- Technical Specification For :- RS-1,2,3 & 4

(Qty.-5 Nos)

S.N.	Particulars	JSW Requirement	Remark

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1	**General Technical Spec.**		
1.1	Shutter Type (MSWRS)	Motorized Single Wall Rolling Shutters	
1.2	**RS-1 :-** Size **7M (W) X 4.5 M (H)** Along E-Row (14-14A) - **Truck Passage.**	1 Nos. (Full Slat Curtain)	Yes
1.3	**RS-2 :-** Size **4.0M (W) X4.0M (H)** Along F-Row (14-14A) - **Truck Passage.**	1 Nos. (Full Slat Curtain **with wicket Door**)	Yes
1.4	**RS-3 :-** Size **4M (W) X 6.0M (H)** Along E-Row (4-5), (6-7) - **RTO Room**	2 Nos. (Full Slat Curtain)	Yes
1.5	**RS-4 :-** Size **4M (W) X 4.5 M (H)** Along E-Row (15A-16)- **Paint Room**	1 Nos. (Full Slat Curtain **with wicket Door**)	Yes
1.6	Opening Direction	Vertical	Yes
1.7	Mounting Location	Inside	Yes
1.8	Mounting Surface	**RS-1 & 4** on Brick Wall & **RS-2 & 3** on Steel Frame Structure	Yes
1.9	Wind Resistance	Class 3- Classification as per EN 12424	Class 1
1.10	Air permeability	Class 2- Classification as per EN 12426	NA
1.11	Resistance to water penetration	Class 3- Classification as per EN 12425	NA
2	**Details of Door Construction & Structure**		
2.1	Slat Material	Galvalume Steel with AZ-100	GI
2.2	Slat Profile & Height	**Curved** & supplier has to mentioned the height (Aprox.-88 mm)	Curved slats
2.3	Slat Thickness	**1.0 mm** thickness	Yes
2.4	Curtain Weight (kg / m2)	**Supplier has to specify** (Approx. 13 Kg/M2)	Yes
2.5	Curtain Material	Single Skinned Steel	Yes
2.6	Curtain Surface	Galvalume Steel should be free from all surface defect. If surface defect found then it will rejected. No Branding on curtain material. (No Logo of company)	GI steel with one coat of grey epoxy primer by spray
2.7	Side Guide Frame	Galvanized Steel side guide **100 mm wide** & **2.00 mm thickness** with seal & wind lock Hooks.	MS Guides
2.8	Wind lock Storm Anchors	Require	Ok

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2.9	Bottom Profile	MS bottom plate **2.5 mm thick** with MS angle stiffener 35 x 35 x 5 mm welded to full opening width for reinforcement. L-shaped bottom profile interlocked with bottom most slat section with Rubber sealing at bottom	Yes
2.10	Hood & Motor Cover	Galvanized Steel Hood & Motor Cover for Dust Protection (0.80 mm thickness)	Yes
2.11	Roller Shaft & Bracket	As per Supplier's Design :- Torsion spring shaft with lateral load-bearing cables & Heavy duty steel Bracket with one coat of epoxy primer & one coat of Paint. **(Supplier has to specify the size & thickness)**	Will give after PO
2.12	Sealing	EPDM Rubber Seal at Bottom.	Yes
2.13	Wicket Door	**For RS-2 & 4** & Others are not Require	Yes
2.14	Vision Panel	**As per Design According to width** (At 1.2 m height from bottom), Glazing: Rectangular glazing constructed with single or double glazed panels. Approx. Vision Panel Size: 600-650 mm (W) x 300-350 mm (H)	NO
2.15	Surface Finish	One coat of primer and one coat of paint on side guides, Bottom profiles and covers.	Yes
3	**Details of Automation & Accessories**		
3.1	Control Type	Pulse Control with dead man control	Yes
3.2	Drive Unit	Direct Drive easy to mount with in built Limit Switches. **Supplier has to specify the drive details.**	Yes
3.3	No of cycle per Day (In 24 Hrs.)	10-15 cycles in a Day	Yes
3.4	Operating Switch	Three Station Push Button :- **2 Nos.** for Inside & Outside	Yes
3.5	Emergency Operation	Manual Hand Chain	Yes
3.6	Opening/Closing Speed	Up to **0.1m/sec,**	Yes
3.7	Safety Catch for door (Torsion spring breakage protection device)	Breaking device to avoid falling by self wt.	Yes
3.8	Safety Side trap guard	For Cable & Manual intervention.	Yes
3.9	Mechanical / Operational / Safety Accessories	Hood, Shaft & Motor Cover: GI Sheet of Min. 0.8mm thick	Yes

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		Bottom Rubber Sealing	Yes
		Fixed Bollards (4 Nos.per Door of 800 mm height)	Yes
		Reflective/Indicative strips on guides/bottom profile	Yes
		Key Switch	Yes
		Flashing Light with Alarm	Yes
		Traffic Light, Red for Closed & Green for Open on both side (Inside &Outside)	Yes
4	**Spare List for Rolling Shutter**		
4.1	Controller box (including Card and Push button)	2 No	Yes
4.2	Special cable (From Controller to sensor)	1 Set	Yes

Sr. No.-11 :- Technical Specification For :- RS-5 (Qty.-3 Nos)

S.N.	Particulars	JSW Requirement	Remark
1	**General Technical Spec.**		
1.1	Shutter Type (MSW-Grill-RS-5)	Motorized Single Wall Rolling Shutters with partly Grill Curtain	Yes
1.2	**RS-5 :-** Size **5M (W) X 4.5M (H)** Along E-Row (10-10A), (10A-11), (11-11A) - **X^{mer} Room.**	3 Nos. (Partly Grill curtain in Honey comb Pattern at 1.2 m, of height 1.0 m)	Yes
1.3	Opening Direction	Vertical	Yes
1.4	Mounting Location	Inside	Yes
1.5	Mounting Surface	Steel Frame .	Yes
1.6	Wind Resistance	Class 3- Classification as per EN 12424	Class 1
1.7	Air permeability	Class 2- Classification as per EN 12426	NA
1.8	Resistance to water penetration	Class 3- Classification as per EN 12425	NA
2	**Details of Door Construction & Structure**		
2.1	Slat Material	Galvalume Steel with AZ-100	GI

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2.2	Slat Profile & Height	**Curved** & supplier has to mentioned the height (Aprox.-88 mm)	Curved
2.3	Slat Thickness	**1.0 mm** thickness	Yes
2.4	Curtain Weight (kg / m2)	**Supplier has to specify** (Approx. 13 Kg/M2)	Yes
2.5	Curtain Material	Single Skinned Steel	Yes
2.6	Curtain Surface	Galvalume Steel should be free from all surface defect. If surface defect found then it will rejected. No Branding on curtain material. (No Logo of company)	GI steel with one coat of grey epoxy primer by spray
2.7	Side Guide Frame	Galvanized Steel side guide **100 mm wide** & **2.00 mm thickness** with seal & wind lock Hooks.	MS
2.8	Wind lock Storm Anchors	Require	Ok
2.9	Bottom Profile	MS bottom plate **2.5 mm thick** with MS angle stiffener 35 x 35 x 5 mm welded to full opening width for reinforcement. L-shaped bottom profile interlocked with bottom most slat section with Rubber sealing at bottom.	Yes
2.10	Hood & Motor Cover	Galvanized Steel Hood & Motor Cover for Dust Protection (0.80 mm thickness)	Yes
2.11	Roller Shaft & Bracket	As per Supplier's Design :- Torsion spring shaft with lateral load-bearing cables & Heavy duty steel Bracket with one coat of epoxy primer & one coat of Paint. **(Supplier has to specify the size & thickness)**	Yes
2.12	Sealing	EPDM Rubber Seal at Bottom.	Yes
2.13	Wicket Door	**Not Required**	
2.14	Vision Panel	Partly Grill curtain in Honey comb Pattern at 1.2 m, **of height 1.0 m**	Yes
2.15	Surface Finish	One coat of primer and one coat of paint on side guides, Bottom profiles, Grill curtain and covers	Yes
3	**Details of Automation & Accessories**		
3.1	Control Type	Pulse Control with dead man control	Yes
3.2	Drive Unit	Direct Drive easy to mount with in built Limit Switches. **Supplier has to specify the drive details.**	Yes

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3.3	No of cycle per Day (In 24 Hrs.)	10-15 cycles in a Day	Yes
3.4	Operating Switch	Three Station Push Button :- **2 Nos.** for Inside & Outside	Yes
3.5	Emergency Operation	Manual Hand Chain	Yes
3.6	Opening/Closing Speed	Up to **0.1m/sec,**	Yes
3.7	Safety Catch for door (Torsion spring breakage protection device)	Breaking device to avoid falling by self wt.	Yes
3.8	Safety Side trap guard	For Cable & Manual intervention.	Yes
3.9	Mechanical / Operational / Safety Accessories	Hood, Shaft & Motor Cover: GI Sheet of Min. 0.8mm thick	Yes
		Bottom Rubber Sealing	Yes
		Fixed Bollards (4 Nos.per Door of 800 mm height)	Yes
		Reflective/Indicative strips on guides/bottom profile	Yes
		Key Switch	Yes
		Flashing Light with Alarm	Yes
		Traffic Light, Red for Closed & Green for Open on both side (Inside &Outside)	Yes
4	**Spare List for Doors**		
4.1	Controller box (including Card and Push button)	2 No	Yes
4.2	Special cable (From Controller to sensor)	1 Set	Yes

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Installation & Service Scope :-

S.N.	Particulars	JSW Requirement	Remark
1	Door Installation & commissioning	Supplier's Scope	Yes
2	All tools and tackles including welding machine , welding rod & other consumables etc. required for installation .	Supplier's Scope	Yes
3	Unloading & Storage at site	JSW Scope	Yes
4	Shifting of the material to installation location	Supplier's Scope	Yes
5	Scaffolding / Mobile crane / Farrana for Gate Material shifting & Installation (if required)	Supplier's Scope	Yes
6	Delivery time	Supplier to Specify (6 to 8 week)	12-14 weeks for Sectional door and 6 -7 weeks for Rolling Shutter
7	Availability of Service engineer in case of requirement	Supplier to Specify (With in 24 hrs.)	Yes
8	Installation Time / Door	Supplier to Specify (Approx. 1 to 2 Days/Door)	Yes
9	Warranty with Replacement of Faulty parts	Supplier to Specify (Min. 1-Year)	Yes

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JSW General Guideline for the Job Execution:-

A	Gate pass requirements for workers:
1	ID Proof
2	ESIC policy
3	Passport size photograph
4	For operators & drivers of crane, Farrana, man-lift etc special category valid licenses copy
5	Safety training is mandatory before deputing the workers/staff on site for actual work.
B	**Gate pass requirements for Crane, Farrana, man-lift, Truck:**
1	Valid RTO registration including Fitness certificate
2	RC book if applicable
3	Test certificate (form No. 11) for all lifting machinery
4	PUC
5	Insurance certificate
C	**General Safety Rules as per JSW Standard.**
1	Contractors has to provide the PPEs to their workers/supervisor on their own cost. PPEs includes Helmet, Safety Goggles, Fluorescent Jacket of orange color, Safety Shoes, Hand gloves, Ear Plugs, Respirator Mask & Safety Belts etc.)
2	4-Nos of good condition PPEs like Helmet, Safety Goggles, Fluorescent Jacket of orange color & Safety Shoes are compulsory to every persons working at site including worker & supervisors. And other PEEs has to be used as per job requirement.
3	One Safety officer is require for each 20 nos. of workmen & should be available at site

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	throughout the working hours.
4	Gas Cutting set should have proper trolley arrangement with flash back arrester, pressure gauges, spark lighter and good condition hoses. All empty cylinders should in vertical position and properly placed at site.
5	All Welding machines should have ELCB with good condition of earthing & power cable.
6	Gas cutting, Grinding, Cutting & Welding jobs without PPEs are not acceptable. Proper use of PPEs like safety goggles, welding screen & hand gloves are mandatory.
7	Joint less electric cables with proper three pin plug top for each equipment is necessary at site.
8	Proper Scaffolding of steel pipes as per JSW Standard with necessary supports, clamping arrangement, working platform & safety railings for working at height above 1.8 meter is mandatory.
9	Use of good condition safety belt with proper anchoring / lifeline is compulsory for working at Height above 1.8 M.
10	PTW (Permit to Work) with proper certificates and risk assessment is mandatory for all kinds of job on daily basis and it will be valid for 8 Hours. In case of any extension of work above 8 hours, permit holder has to extend the PTW for extended hours by getting the signature of authorizer.
11	One filled hard copy of PTW (White Color) to be displayed at site with signature of all concern.
12	Everyday safety talks to be conducted at site before starting the job. And mention the Name & signature of all workmen who has attended the safety talk at the back side of PTW
13	PTW closing is compulsory after completion of work every day.
14	Shifting of material from one place to other with the help of farrana in hanging position inside the company premises is completely banned. Farrana will be used only for loading, unloading & erection of material. And for shifting of material truck, trailer or trolley to be used.
15	Proper lifting tool & tackles with test certificate is necessary.
16	Non-compliance/Violation of any of the above rule will attract penalty of Rs 500 /- for single instance.

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